



Market Assessment of the Enhanced Rock Weathering (ERW) Industry for WeAct: Strategic Entry Evaluation and Growth Opportunities

Prepared for **WeAct.com.au**

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1. Introduction to WeAct

WeAct is an Australian carbon project developer and trading house that builds, operates, and finances

nature- and technology-based climate projects. Founded in 2009, the firm focuses on delivering measurable climate outcomes alongside sustainable development benefits for local communities.

WeAct operates through three core pillars: **We Build** (owning and operating carbon projects), **We Partner** (providing technical and operational support for land and asset owners), and **We Trade** (a global carbon trading desk that distributes verified credits and reinvests proceeds into climate solutions).

The company has operations across multiple countries and emphasises community engagement, Free, Prior & Informed Consent principles, and transparent measurement, reporting, and verification processes to ensure high-integrity carbon outcomes.

2. Executive Summary

WeAct is a leading carbon project developer seeking growth opportunities in emerging carbon removal markets. The Enhanced Rock Weathering (ERW) industry has gained attention for its potential to sequester CO₂ and generate high-integrity carbon credits.

ERW is an emerging market with technical variability, site-dependent performance, evolving MRV standards, and operational complexities. WeAct lacks detailed market intelligence to evaluate feasibility and strategic fit.

Should WeAct enter the ERW market, and if so, how can it leverage its existing capabilities to maximize impact while managing risks?

WeAct should pursue a **phased entry into ERW**, starting with pilot projects in high-suitability regions. Leveraging its in-house science team, trading expertise, and partnerships, WeAct can scale operations, ensure rigorous MRV compliance, and position itself as an early mover in this growing carbon removal market. A structured risk management, implementation roadmap, and change management plan will ensure strategic alignment, operational efficiency, and sustainable growth.

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4. Our Understanding of the Problem

WeAct is exploring potential expansion into the Enhanced Rock Weathering (ERW) industry as part of its broader strategy to diversify its portfolio of high-integrity carbon removal and climate solutions. ERW is an emerging carbon dioxide removal (CDR) pathway with growing academic validation, increasing global investment, and early commercial activity driven by voluntary carbon market demand.

However, the ERW market remains fragmented, technologically variable, and characterised by uncertainties

relating to scalability, measurement, reporting and verification (MRV), cost structures, policy frameworks, and long-term carbon credit integrity. Competitors range from early-stage startups to vertically integrated carbon removal developers, each with differing business models and levels of scientific robustness.

To make an informed entry decision, WeAct requires a structured and data-driven market assessment that clarifies the size, maturity, and growth trajectory of the ERW industry; evaluates competitive dynamics and business models; identifies key risks and operational requirements; and defines viable market entry pathways that align with WeAct's capabilities, partnerships, and existing project development strengths.

This consulting engagement aims to deliver a comprehensive understanding of the ERW ecosystem so that WeAct can determine whether entering this market is strategically, financially, and operationally attractive.

5. Problem Statement

WeAct seeks to determine whether entering the Enhanced Rock Weathering (ERW) carbon removal market is a viable and strategically aligned growth opportunity.

The company requires clarity on the market's size, maturity, competitive landscape, technological feasibility, MRV integrity, and commercial viability. Currently, WeAct lacks the detailed market intelligence needed to evaluate ERW's potential fit with its existing project development capabilities, partnership model, and long-term carbon strategy.

This report aims to address this gap by assessing the ERW industry and identifying whether WeAct should enter the market, and if so, through which strategic pathway.

6. Scope and Out-of-Scope

In-Scope:

- Comprehensive market assessment of the Enhanced Rock Weathering (ERW) industry, including market size, growth trends, and maturity.
- Analysis of competitive landscape, including key players, business models, and partnerships.

- Evaluation of technological feasibility, operational requirements, and measurement, reporting, and verification (MRV) practices.
- Identification of strategic entry options aligned with WeAct’s capabilities and portfolio.
- Preliminary assessment of risks, opportunities, and potential barriers to entry.

Out-of-Scope:

- Detailed financial modeling or investment due diligence for specific ERW projects.
- Implementation of any ERW projects or pilot programs.
- In-depth regulatory compliance reviews outside publicly available policies and frameworks.
- Extensive field-level technical trials or lab experiments.

This scope ensures focus on strategic market insights necessary for WeAct to make an informed entry decision while excluding operational execution or detailed financial commitments.

7. Methodology

1. PESTEL Analysis

We conducted a PESTEL analysis to evaluate the macro-environmental factors affecting the ERW industry. This includes assessing Political, Economic, Social, Technological, Environmental, and Legal dimensions that could impact market entry, regulatory compliance, and scalability.

2. Ansoff Matrix Analysis

An Ansoff Matrix was used to explore potential growth strategies for WeAct within the ERW market. This includes assessing options for market penetration, product development, market development, and diversification to determine the most strategically aligned growth pathway.

3. Porter’s Five Forces Analysis

To understand competitive dynamics, we applied Porter’s Five Forces framework. This examined the bargaining power of suppliers and buyers, threat of new entrants, threat of substitutes, and competitive

rivalry to evaluate the attractiveness and profitability potential of the ERW market.

4. SWOT Analysis for WeAct

A SWOT analysis was conducted to identify WeAct's internal strengths and weaknesses, as well as external opportunities and threats related to ERW market entry. This provides context for aligning the company's capabilities with strategic options.

5. VRIO Analysis for WeAct's In-House Science Team

We applied the VRIO framework to evaluate the value, rarity, inimitability, and organizational support of WeAct's in-house science team. This assesses whether internal scientific capabilities provide a sustainable competitive advantage in ERW project development.

6. Data Collection & Analysis

Primary and secondary data sources were reviewed, including industry reports, academic research, market databases, and public carbon project information. Key metrics such as market size, growth rate, cost structures, and carbon credit potential were extracted and analyzed.

7. Synthesis & Recommendation Development

Insights from the above analyses were synthesized to identify strategic entry options, evaluate feasibility, highlight risks, and form actionable recommendations for WeAct's potential entry into the ERW market.

8. Hypothesis

WeAct can successfully enter the Enhanced Rock Weathering (ERW) market if the industry demonstrates sufficient commercial viability, technological feasibility, and alignment with WeAct's existing capabilities, partnerships, and in-house scientific expertise.

Entry into ERW is likely to create strategic growth opportunities by expanding WeAct's carbon removal portfolio, generating high-integrity carbon credits, and strengthening its position in emerging climate solution markets.

9. Data

Industry Reports & Market Studies

- Global and regional ERW market reports
- Carbon removal and voluntary carbon credit market analyses
- Sustainability and climate solution trend reports

Academic Research & Scientific Publications

- Peer-reviewed studies on ERW efficacy, scalability, and carbon sequestration potential
- Technical assessments of mineral weathering processes
- Environmental impact and lifecycle analyses

Competitor & Benchmarking Data

- Public information on ERW startups and carbon removal companies
- Business models, partnerships, and project case studies
- Carbon credit generation performance metrics

Government & Policy Sources

- National and international carbon regulations and guidelines
- Climate policy frameworks affecting carbon removal adoption

- Incentives, subsidies, or research grants available for ERW projects

Market Databases & Trade Data

- Voluntary carbon credit marketplaces
- Transaction and pricing trends for carbon credits
- Investment activity and funding rounds in ERW and CDR technologies

Internal WeAct Data & Capabilities

- Existing project development expertise
- In-house scientific and operational capabilities
- Partnerships, past project performance, and carbon portfolio metrics

Expert Interviews & Stakeholder Consultations

- Discussions with ERW researchers and practitioners
 - Consultations with market analysts and climate solution investors
 - Insights from potential customers and carbon credit buyers
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10. Data Analysis

Market Size & Growth Trends

- Quantified current ERW market size and projected growth over the next 5–10 years.
- Identified emerging geographies and sectors driving adoption of ERW carbon credits.

Competitive Landscape Analysis

- Compared ERW players based on business models, project scale, partnerships, and scientific credibility.
- Evaluated barriers to entry and differentiation opportunities for WeAct.

Technological Feasibility & MRV Assessment

- Examined scientific studies on mineral weathering efficiency, scalability, and carbon sequestration potential.
- Reviewed measurement, reporting, and verification (MRV) methods used by existing ERW projects to assess reliability and compliance with voluntary carbon standards.

Strategic Framework Analyses

- **PESTEL:** Identified macro-level risks and opportunities across political, economic, social, technological, environmental, and legal dimensions.
- **Ansoff Matrix:** Explored growth strategies suitable for WeAct, including market penetration, product development, and diversification.
- **Porter's Five Forces:** Evaluated industry attractiveness by analyzing supplier/buyer power, threats of new entrants/substitutes, and competitive rivalry.
- **SWOT & VRIO:** Assessed WeAct's internal strengths, weaknesses, opportunities, and threats;

evaluated the in-house science team for sustainable competitive advantage.

Financial & Cost Structure Insights

- Analyzed estimated cost per ton of CO₂ removal via ERW versus other carbon removal pathways.
 - Evaluated revenue potential from carbon credits and associated market premiums.
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11. Insights from Data

1. Market & Scientific Insights

- **Market Size & Growth:** The global ERW market is emerging (~USD 1.28B in 2024) with projected CAGR of ~19.7% to reach USD 6.14B by 2033 ([DataIntello, 2025](#)).
- **Scientific Efficacy:** Field studies indicate ERW can increase soil organic carbon by 3–8% depending on soil, climate, and rock type ([PubMed, 2023–2025](#)).
- **Technological Feasibility:** Basalt and other silicate rocks are the main feedstock. Scalability depends on mineral sourcing, site selection, and operational controls.

Why Demand for Carbon Offsets Could Surge

Carbon offsets—which play an important role in achieving global climate goals—could see staggering demand growth of **15-fold** by 2030.

15X



The market for carbon offsets could be worth upward of \$50 billion in 2030.

Soaring Demand Projected

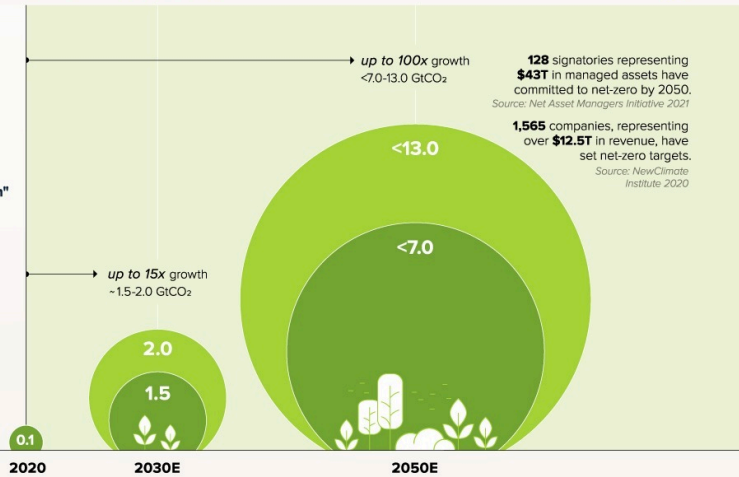
Voluntary carbon offsets, gigatons of CO₂ per year (GtCO₂)



NGFS scenarios*



NGFS "immediate action"
1.5°C pathway with carbon-dioxide removal*



Today, just **21.5%** of emissions are covered by carbon pricing initiatives, setting the stage for significant growth in voluntary carbon markets.

Source: World Bank 2021



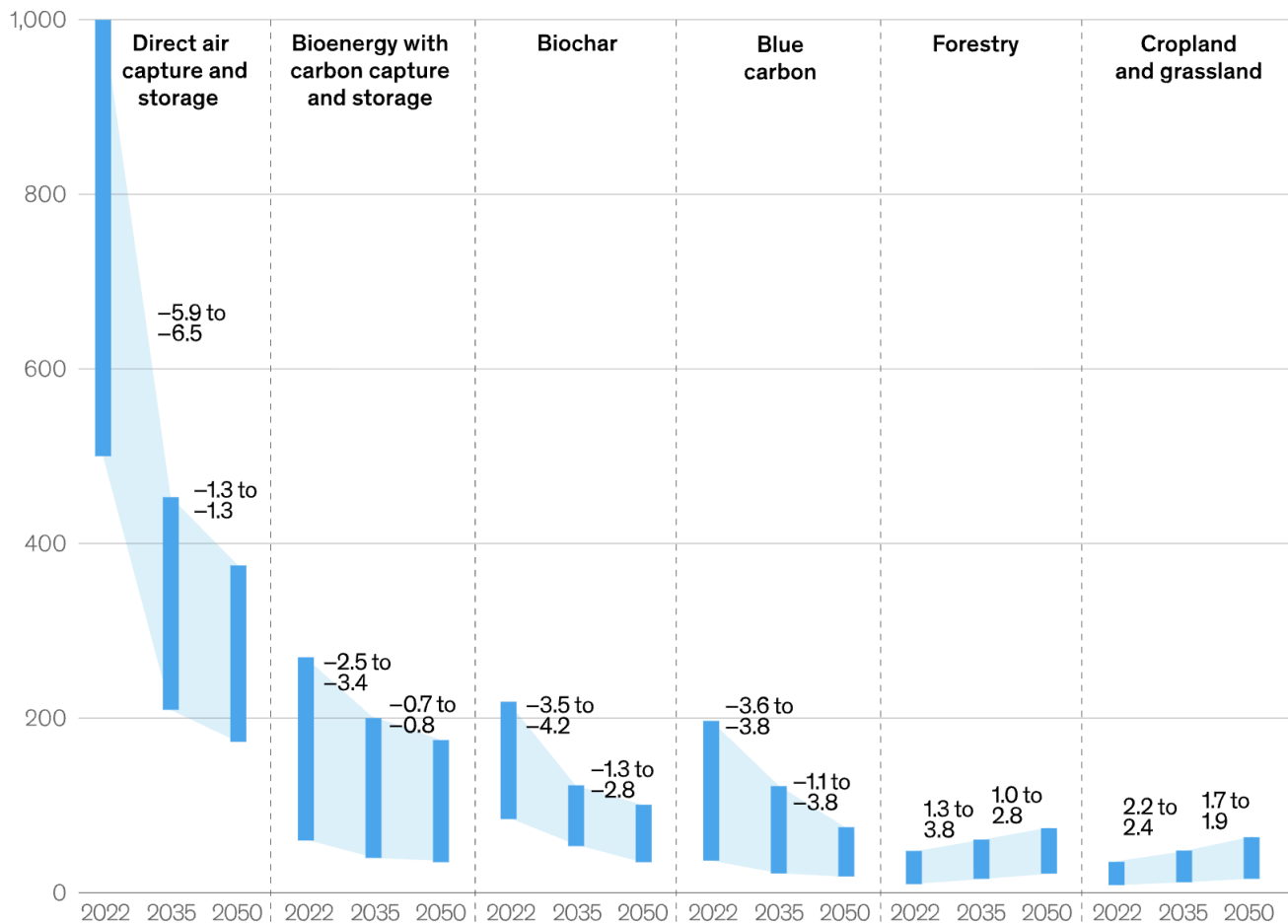
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CARBONSTREAMING.COM

*NGFS= Network for Greening the Financial System. These amounts reflect demand based on carbon-dioxide removal and sequestration requirements under the NGFS's 1.5°C and 2.0°C scenarios. Both amounts reflect an assumption that all carbon-dioxide removal and sequestration results from carbon credits purchased on the voluntary market (whereas some removal and sequestration will result from carbon credits purchased in compliance markets and some will result from efforts other than carbon-offsetting projects).
Source: McKinsey & Company, "A blueprint for scaling voluntary carbon markets to meet the challenge," Jan 2021



Levelized cost, \$ per metric ton of CO₂

x.x to x.x = Estimated CAGR across time period, %



Source: McKinsey analysis based on TRAILS and Nature Analytics land-use modeling and technology-specific carbon management service line cost models

McKinsey & Company

2. Scientific Evidence: Carbon Sequestration & Soil Health

- A recent meta-analysis synthesising 74 field studies found that applying crushed silicate rocks (ERW) increased soil organic carbon (SOC) by up to **3.8%**, mineral-associated organic carbon by **~6.1%**, and particulate organic carbon by **~7.5%** (depending on soil type, climate, and application dosage). [PubMed](#)
- Results vary significantly by geographic and climatic context — positive SOC accumulation was more pronounced in **low-latitude (warm and humid) regions** ($\approx 40^\circ \text{N}$ – 30°S), while some high-latitude or arid regions showed negligible gains or potential decreases. [PubMed](#)
- A 2-year field experiment in a tropical rubber plantation found significant increases in both soil organic carbon and inorganic carbon (bicarbonate) after ERW with silicate (wollastonite) addition — in some cases, SOC increased four to eight times more than inorganic carbon levels. [PubMed](#)

- Recent 2025 field work in a temperate forest plantation (larch plantation, northeastern China) observed reductions in soil CO₂ flux by ~16.5% in the first year after ERW application — suggesting improved soil carbon stability. Over a longer timeframe, tree biomass also trended upward (though not statistically significant), hinting at vegetation-mediated carbon sequestration in addition to soil storage. [ScienceDirect+2Chinese Academy of Sciences+2](#)

Implication: ERW has demonstrated credible carbon sequestration potential under favorable conditions, and also offers co-benefits like soil health improvement and possibly enhanced vegetation growth — attractive for a carbon project developer with existing nature-based solutions expertise (like WeAct).

3. Technological Feasibility, Controls & Variability

- Modelling studies of mineral weathering (e.g. using forsterite as proxy mineral) show that under ideal conditions (good soil drainage, sufficient CO₂ supply via microbial or root respiration), a one-time application of ~16 kg/m² could lead to complete rock weathering within 5 years, corresponding to ~**2.3 kg CO₂ / m² / year** removal — though rates vary significantly with site conditions. [PubMed+1](#)
- Rock type matters: Silicate rocks like basalt dominate ERW feedstock globally — due to availability, reactivity, and lower environmental risk relative to ultramafic rocks. Basalt-based ERW accounts for a large share of market volume today. [Market Intel+1](#)
- Variability & site dependency remain key challenges. Soil type, climate (temperature, moisture), crop type / vegetation, and application dosage all influence weathering rates and carbon gains. [PubMed+2PubMed+2](#)
- Lifecycle and energy costs — notably mining, crushing, transport and application of rock — remain non-trivial. According to assessments from the Massachusetts Institute of Technology (MIT) Climate Portal, energy-intensive rock processing and logistics can offset gains if not managed carefully; large land-use and scalability challenges exist. [MIT Climate Portal+1](#)

Implication: ERW is technically feasible — but net carbon removal and commercial viability require careful selection of site, rock type, and operational parameters. For WeAct, leveraging supply-chain expertise and project-management capabilities will be critical to manage variability and reduce costs.

4. Market & Commercial Viability: Carbon Credit Demand & Verification Needs

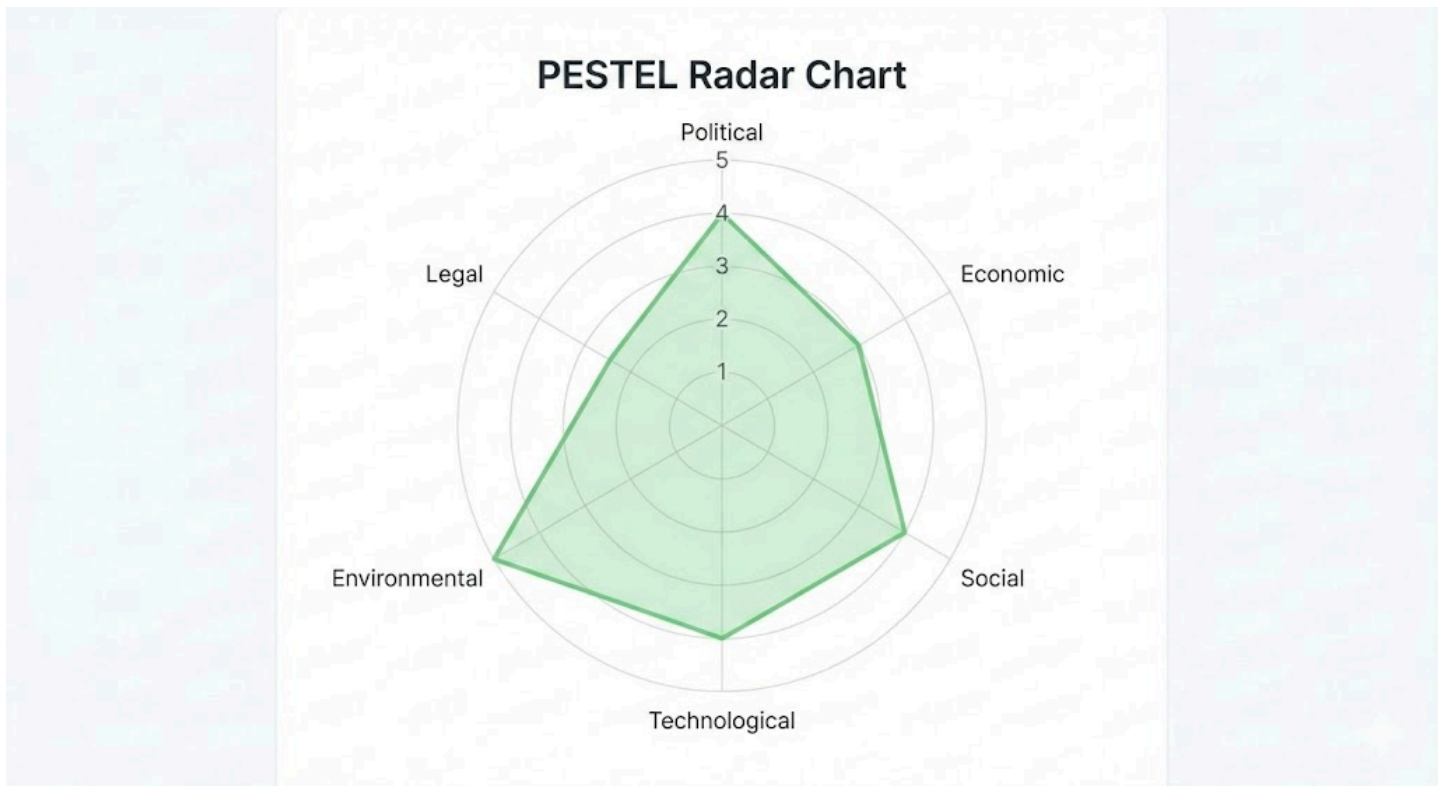
- As demand for carbon removal credits grows (owing to corporate net-zero pledges, regulatory pressure, and voluntary carbon markets), ERW is emerging as a potentially scalable CDR pathway.

- The market for ERW verification — i.e. systems and services that measure, report, and verify (MRV) carbon removal — is projected to grow substantially: one market outlook estimates the ERW verification market at **USD 156 million in 2024**, with a projected CAGR of ~21.3% through 2033, potentially reaching **~USD 1.11 billion**. [Growth Market Reports](#)
- This underscores the increasing importance of robust science-based MRV frameworks — a necessity for credible carbon credits and investor confidence. Research initiatives are already exploring mass-balance soil-based monitoring methods to provide reliable in-situ estimates of CO₂ removal. [arXiv+1](#)

Implication: For a company like WeAct — which has an established “Build-Operate-Trade” model and trading desk — entering ERW with a strong MRV infrastructure offers a pathway to generate high-integrity credits and meet growing demand, potentially opening new revenue streams.

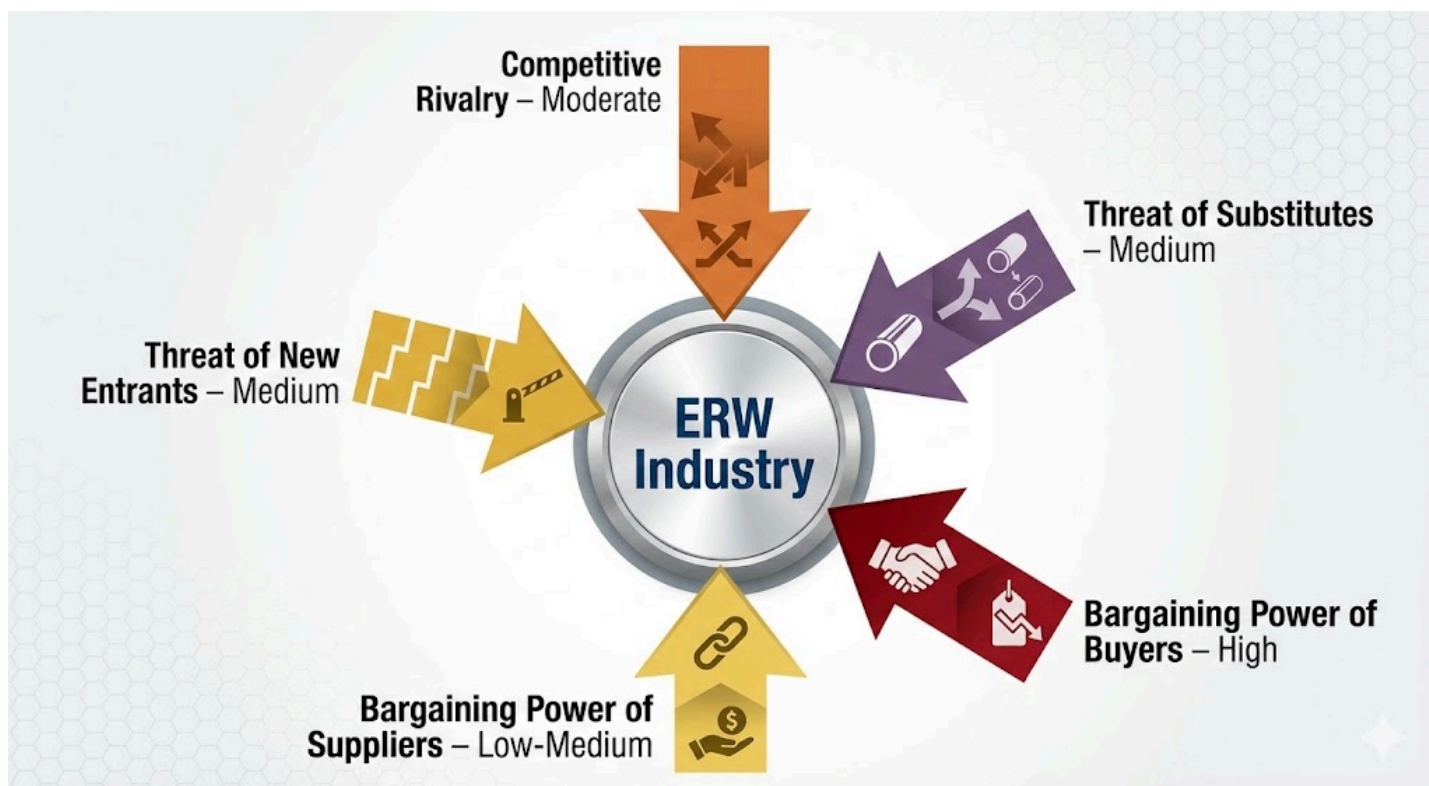
5. PESTEL Analysis

Factor	Key Insights
Political	Supportive climate policies in Australia and select regions; potential subsidies for carbon removal projects.
Economic	High upfront CAPEX for ERW projects; long-term carbon credit revenue potential.
Social	Positive impact on soil health and community sustainability; engagement required to ensure consent and acceptance.
Technological	Advances in MRV, monitoring, and mineral application enhance feasibility.
Environmental	Co-benefits include soil enrichment and possible vegetation growth; minimal ecological disruption if applied responsibly.
Legal	Carbon credit regulations evolving; compliance with MRV standards essential.



6. Porter's Five Forces

- **Competitive Rivalry:** Moderate; early movers gaining credibility.
- **Threat of New Entrants:** Medium; requires scientific expertise and operational capacity.
- **Bargaining Power of Suppliers:** Low-to-medium; rock sourcing and transport are manageable with partnerships.
- **Bargaining Power of Buyers:** High; corporate buyers demand high-integrity carbon credits.
- **Threat of Substitutes:** Medium; other CDR pathways (afforestation, DAC) may compete.



7. SWOT Analysis (WeAct in ERW Context)

Strengths

- Established carbon project portfolio
- Experienced in carbon trading
- Strong in-house science team

Weaknesses

- Limited ERW-specific operational experience
- Need for specialized MRV for ERW
- Supply chain/logistics for minerals need development

Opportunities

- First-mover advantage in ERW credits
- Growing voluntary carbon market
- Potential for high-integrity carbon credits

Threats

- Technological/regulatory uncertainties
- Market adoption lag or MRV inconsistency
- Competitors with early credibility

SWOT Analysis: ERW Carbon Project

Strengths	Weaknesses
- Established carbon project portfolio - Experienced in carbon trading - Strong in-house science team	- Limited ERW-specific operational experience - MRV for ERW needs development - Supply chain/logistics for minerals need development
Opportunities	Threats
- First-mover advantage in ERW credits - Growing voluntary carbon market - Potential for high-integrity carbon credits	- Technological/regulatory uncertainties - Market adoption lag - Competitors with early credibility

8. VRIO Analysis (In-House Science Team)

Resource	Value	Rarity	Inimitability	Organization	Conclusion
Science team	Yes	High	High	Yes	Sustainable competitive advantage
MRV expertise	Yes	Medium	Medium	Yes	Temporary advantage

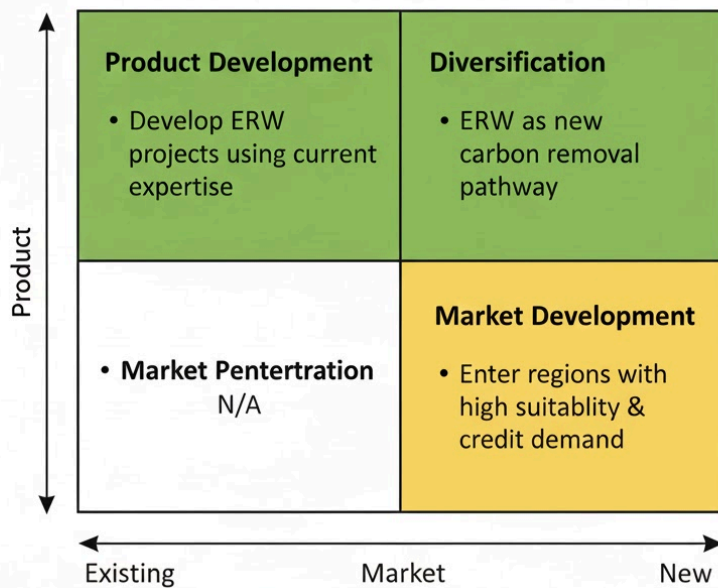
Implication: WeAct's science team can support credible ERW projects and maintain competitive edge in MRV and project design.

7. Ansoff Matrix Insights

- Market Penetration:** Limited immediate options; ERW is a new market.

- **Product Development:** Develop ERW projects leveraging existing carbon development expertise.
- **Market Development:** Enter regions with high ERW suitability and corporate demand for credits.
- **Diversification:** ERW represents a new carbon removal pathway complementing WeAct's portfolio.

Ansoff Matrix: ERW Carbon Strategy Options



8. Key Strategic Insights

- ERW is scientifically credible and commercially emerging.
- Market entry requires leveraging WeAct's in-house science team, trading desk, and project development experience.
- MRV standardization, operational control, and partnerships are critical to ensure success.

- Early entry offers strategic advantage in a growing voluntary carbon market.
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12. Recommendation

Based on the market assessment, technological feasibility, and internal capability evaluation, we recommend that **WeAct pursue a strategic entry into the Enhanced Rock Weathering (ERW) market.**

Key Recommendations:

1. Phased Market Entry

- Begin with **pilot ERW projects** in regions with high soil suitability and supportive policy frameworks.
- Scale operations gradually, leveraging learnings from initial projects to refine processes, MRV, and operational workflows.

2. Leverage Internal Capabilities

- Utilize WeAct's **in-house science team** to ensure rigorous MRV, project design, and carbon credit integrity.
- Integrate ERW projects with existing trading desk operations to capture early revenue from verified carbon credits.

3. Strategic Partnerships

- Partner with **mineral suppliers, local landholders, and research institutions** to optimize operational efficiency and reduce risks.
- Collaborate with MRV technology providers to ensure compliance with voluntary carbon standards.

4. Risk Management & MRV Standardization

- Implement robust **monitoring, reporting, and verification frameworks** to ensure credibility in carbon credit markets.

- Track site-specific conditions, operational performance, and regulatory updates to manage technical and compliance risks.

5. Growth & Diversification Strategy

- Use **Ansoff Matrix insights**: develop new ERW projects (product development) and diversify the carbon portfolio to strengthen market position.
- Explore international expansion in high-demand markets once internal capabilities and partnerships are established.

Conclusion:

Entering the ERW market aligns with WeAct’s mission to deliver high-integrity carbon projects while leveraging its scientific, operational, and trading strengths. A phased, risk-managed approach ensures scalability, mitigates uncertainties, and positions WeAct as an early mover in an emerging carbon removal industry.

13. Risks and Risk Management Plan

Entering the Enhanced Rock Weathering (ERW) market involves multiple risks. Below is a summary of key risks and proposed mitigation strategies:

Risk ID	Date Raised	Risk Description	Likelihood of Occurring	Impact	Overall Severity	Owner	Mitigation Strategy
R1	02-Dec-2025	Variability in carbon sequestration due to soil, climate, or rock type	Medium	High	High	Science Team	Conduct pilot projects; use site-specific modeling; optimize rock type and application rates

R2	02-Dec-2025	Inconsistent MRV compliance impacting carbon credit integrity	Medium	High	High	Project Management / Science Team	Develop robust MRV framework; partner with certified MRV providers; implement audits
R3	02-Dec-2025	Regulatory or policy changes affecting ERW carbon credits	Medium	Medium	Medium	Compliance / Legal Team	Monitor policy developments; maintain flexible project design to adapt quickly
R4	02-Dec-2025	Operational & supply chain issues for rock sourcing, transport, and application	Medium	Medium	Medium	Operations Team	Establish supplier partnerships; optimize logistics; implement quality control
R5	02-Dec-2025	Slow adoption of ERW carbon credits by buyers	Medium	Medium	Medium	Trading Desk / Marketing	Engage early with buyers; highlight scientific validation; diversify buyer base
R6	02-Dec-2025	High upfront CAPEX and uncertain revenue timeline	Medium	Medium	Medium	Finance / Strategy Team	Phase project rollout; explore co-financing, grants, or joint ventures

Risk Management Approach:

1. **Identify & Assess:** Continuously monitor potential risks and categorize them by impact and likelihood.
2. **Mitigate & Monitor:** Implement mitigation strategies and establish KPIs to track effectiveness.
3. **Adapt & Respond:** Use pilot projects and phased deployment to adjust operations and reduce exposure to unforeseen challenges.

Conclusion:

A structured risk management plan allows WeAct to enter the ERW market cautiously while minimizing exposure, ensuring credibility in carbon markets, and protecting operational and financial performance.

14. Implementation Roadmap

To ensure a successful entry into the Enhanced Rock Weathering (ERW) market, we recommend a **phased approach** over 3–5 years.

Phase 1: Pilot & Validation

- Identify suitable sites based on soil, climate, and mineral availability.
- Launch small-scale ERW pilot projects to validate carbon sequestration potential.
- Implement **robust MRV protocols** to track performance and generate early verified credits.
- Establish partnerships with local suppliers, research institutions, and landowners.

Phase 2: Initial Market Entry

- Scale ERW operations to additional sites with validated performance.
- Begin selling verified ERW carbon credits through WeAct's existing trading desk.
- Monitor operational efficiency, supply chain logistics, and regulatory compliance.
- Refine project design based on pilot learnings and stakeholder feedback.

Phase 3: Strategic Expansion

- Expand geographically into high-demand markets and regions with favorable policies.
- Strengthen partnerships and explore joint ventures or co-financing opportunities.
- Enhance MRV systems using advanced technology for scalability and transparency.
- Begin integrating ERW projects as a core component of WeAct's diversified carbon portfolio.

Phase 4: Market Leadership & Diversification

- Establish WeAct as a leading ERW carbon credit provider.
 - Explore further innovation, such as combining ERW with complementary CDR pathways.
 - Continuously monitor global carbon markets and adjust strategy to maintain competitiveness.
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15. Change Management Process

Successfully entering the Enhanced Rock Weathering (ERW) market requires organizational alignment and structured change management. The following process ensures smooth adoption across teams, partners, and stakeholders.

Step 1: Prepare the Organization

- **Objective:** Establish awareness and readiness for ERW integration.
- Conduct internal workshops to educate staff on ERW, carbon removal potential, and strategic objectives.
- Align executive leadership on vision, priorities, and resource allocation.

Step 2: Plan the Change

- **Objective:** Define roles, responsibilities, and processes for ERW projects.
- Identify stakeholders across operations, science, trading, and finance.
- Develop clear **project governance structures**, including MRV oversight and reporting channels.
- Prepare risk mitigation strategies for operational, financial, and regulatory uncertainties.

Step 3: Implement the Change

- Launch **pilot ERW projects** and integrate operational workflows with existing project management

systems.

- Train teams on MRV protocols, logistics, and reporting.
- Establish communication channels with external partners (suppliers, landowners, research institutions).

Step 4: Monitor & Adjust

- Track key metrics: carbon sequestration performance, project costs, MRV compliance, and stakeholder satisfaction.
- Use feedback loops to improve processes, address issues, and refine project scale-up strategies.
- Communicate successes and lessons learned to maintain engagement and confidence across teams.

Step 5: Institutionalize Change

- Embed ERW operations into WeAct's standard project development lifecycle.
 - Update policies, training materials, and governance practices to reflect new capabilities.
 - Continuously innovate and optimize ERW practices to maintain competitive advantage and high carbon integrity.
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Thank you.